

H2h Matchup Difference Analysis

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```
library(reticulate)
```

```
## Warning: package 'reticulate' was built under R version 4.4.3
```

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2     4.0.0      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.1
## ✓ purrr      1.2.0
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
from nba_api.stats.endpoints import leaguegamelog
```

```
def get_game_logs(season_year):
```

```
    logs = leaguegamelog.LeagueGameLog(
        season = season_year,
        season_type_all_star = "Regular Season",
        player_or_team_abbreviation = "T",
        timeout = 60
    )
```

```
    df_logs = logs.get_data_frames()[0]
```

```
    return df_logs
```

```
get_regular_season_matchup_stats <- function(season_year){
```

```
  df_logs <- py$get_game_logs(season_year)
```

```
  abbrev_lookup <- df_logs %>%  
    select(Team_Abbreviation, Team_Name) %>%  
    distinct() %>%  
    rename(  
      opp_abbreviation = Team_Abbreviation,  
      opp_team_name = Team_Name  
    )
```

```
  matchup_stats <- df_logs %>%  
    select(  
      Team_Name,  
      Team_Abbreviation,  
      Matchup,  
      WL,  
      PTS,  
      PLUS_MINUS,  
      FGM,  
      FGA,  
      FG3M,  
      FG3A,  
      FTM,  
      FTA,  
      OREB,  
      DREB,  
      REB,  
      AST,  
      STL,  
      BLK,  
      TOV  
    ) %>%  
    mutate(  
      opp_abbreviation = str_sub(Matchup, -3, -1),  
      WIN = ifelse(WL == "W", 1, 0),  
      TS_PCT_GAME = PTS / (2 * (FGA + 0.44 * FTA)),  
      EFG_PCT_GAME = (FGM + 0.5 * FG3M) / FGA,  
      THREE_PT_ATTEMPT_RATE_GAME = FG3A / FGA  
    ) %>%  
    left_join(  
      abbrev_lookup,  
      by = "opp_abbreviation"  
    ) %>%  
    group_by(  
      Team_Name,  
      opp_team_name  
    ) %>%  
    summarize(  
      H2H_GAMES = n(),  
      H2H_WINS = sum(WIN),  
      H2H_WIN_PCT = mean(WIN),  
      H2H_POINT_DIFF = mean(PLUS_MINUS),  
      H2H_PTS = mean(PTS),  
      H2H_TS_PCT = mean(TS_PCT_GAME, na.rm = TRUE),
```

```

H2H_EFG_PCT = mean(EFG_PCT_GAME, na.rm = TRUE),
H2H_THREE_PT_RATE = mean(THREE_PT_ATTEMPT_RATE_GAME, na.rm = TRUE),
H2H_REB = mean(REB),
H2H_AST = mean(AST),
H2H_TOV = mean(TOV),
.groups = "drop"
) %>%
mutate(season = season_year)

return(matchup_stats)
}
head(get_regular_season_matchup_stats("2015-16"))

```

```

## # A tibble: 6 × 14
##   TEAM_NAME OPP_TEAM_NAME H2H_GAMES H2H_WINS H2H_WIN_PCT H2H_POINT_DIFF H2H_PTS
##   <chr>      <chr>          <int>   <dbl>     <dbl>      <dbl>   <dbl>
## 1 Atlanta H... Boston Celti...     4       3     0.75        7.5    110.
## 2 Atlanta H... Brooklyn Nets     3       2     0.667       13.3    101
## 3 Atlanta H... Charlotte Ho...     4       3     0.75       -1.75    90.5
## 4 Atlanta H... Chicago Bulls     4       4       1        13.8    110.
## 5 Atlanta H... Cleveland Ca...     3       0       0       -9.67    99.7
## 6 Atlanta H... Dallas Maver...     2       2       1         9    105
## # i 7 more variables: H2H_TS_PCT <dbl>, H2H_EFG_PCT <dbl>,
## #   H2H_THREE_PT_RATE <dbl>, H2H_REB <dbl>, H2H_AST <dbl>, H2H_TOV <dbl>,
## #   season <chr>

```

```

years <- c(
  "2015-16", "2016-17", "2017-18", "2018-19", "2019-20",
  "2020-21", "2021-22", "2022-23", "2023-24", "2024-25")

h2h_stats <- map_dfr(years, function(year) {
  Sys.sleep(5)
  get_regular_season_matchup_stats(year)
})

head(h2h_stats)

```

```

## # A tibble: 6 × 14
##   TEAM_NAME OPP_TEAM_NAME H2H_GAMES H2H_WINS H2H_WIN_PCT H2H_POINT_DIFF H2H_PTS
##   <chr>      <chr>          <int>   <dbl>     <dbl>      <dbl>   <dbl>
## 1 Atlanta H... Boston Celti...     4       3     0.75        7.5    110.
## 2 Atlanta H... Brooklyn Nets     3       2     0.667       13.3    101
## 3 Atlanta H... Charlotte Ho...     4       3     0.75       -1.75    90.5
## 4 Atlanta H... Chicago Bulls     4       4       1        13.8    110.
## 5 Atlanta H... Cleveland Ca...     3       0       0       -9.67    99.7
## 6 Atlanta H... Dallas Maver...     2       2       1         9    105
## # i 7 more variables: H2H_TS_PCT <dbl>, H2H_EFG_PCT <dbl>,
## #   H2H_THREE_PT_RATE <dbl>, H2H_REB <dbl>, H2H_AST <dbl>, H2H_TOV <dbl>,
## #   season <chr>

```

```

all_series <- read.csv("all_series.csv")

```

```
series_h2h_df <- all_series %>%
  left_join(
    h2h_stats,
    by = c(
      "season" = "season",
      "TEAM_A" = "TEAM_NAME",
      "TEAM_B" = "OPP_TEAM_NAME"
    )
  ) %>%
  left_join(
    h2h_stats,
    by = c(
      "season" = "season",
      "TEAM_B" = "TEAM_NAME",
      "TEAM_A" = "OPP_TEAM_NAME"
    ),
    suffix = c("_A", "_B")
  )
```

```
head(series_h2h_df)
```

##		SERIES_ID	season	TEAM_A		
## 1	Atlanta Hawks vs Boston Celtics	2015-16		Atlanta Hawks		
## 2	Atlanta Hawks vs Cleveland Cavaliers	2015-16		Atlanta Hawks		
## 3	Charlotte Hornets vs Miami Heat	2015-16		Charlotte Hornets		
## 4	Cleveland Cavaliers vs Detroit Pistons	2015-16		Cleveland Cavaliers		
## 5	Cleveland Cavaliers vs Golden State Warriors	2015-16		Cleveland Cavaliers		
## 6	Cleveland Cavaliers vs Toronto Raptors	2015-16		Cleveland Cavaliers		
##	TEAM_B	TEAM_A_WINS	TEAM_B_WINS	WINNER	H2H_GAMES_A	H2H_WINS_A
## 1	Boston Celtics	4	2	1	4	3
## 2	Cleveland Cavaliers	0	4	0	3	0
## 3	Miami Heat	3	4	0	4	2
## 4	Detroit Pistons	4	0	1	4	1
## 5	Golden State Warriors	4	3	1	2	0
## 6	Toronto Raptors	4	2	1	3	1
##	H2H_WIN_PCT_A	H2H_POINT_DIFF_A	H2H_PTS_A	H2H_TS_PCT_A	H2H_EFG_PCT_A	
## 1	0.7500000	7.500000	110.25000	0.5998403	0.5646531	
## 2	0.0000000	-9.666667	99.66667	0.5031549	0.4592226	
## 3	0.5000000	2.000000	99.25000	0.5399156	0.5092926	
## 4	0.2500000	-1.750000	102.75000	0.5507291	0.5147921	
## 5	0.0000000	-20.000000	90.50000	0.4741638	0.4340396	
## 6	0.3333333	5.333333	106.00000	0.6175113	0.5824544	
##	H2H_THREE_PT_RATE_A	H2H_REB_A	H2H_AST_A	H2H_TOV_A	H2H_GAMES_B	H2H_WINS_B
## 1	0.2913577	41.00000	28.25000	17.25000	4	1
## 2	0.3287328	42.66667	23.33333	14.33333	3	3
## 3	0.3105902	40.75000	21.25000	10.25000	4	2
## 4	0.3010723	42.50000	20.25000	12.50000	4	3
## 5	0.2812714	45.50000	17.00000	13.50000	2	2
## 6	0.3746429	40.33333	22.66667	12.66667	3	2
##	H2H_WIN_PCT_B	H2H_POINT_DIFF_B	H2H_PTS_B	H2H_TS_PCT_B	H2H_EFG_PCT_B	
## 1	0.2500000	-7.500000	102.7500	0.5244228	0.4856816	
## 2	1.0000000	9.666667	109.3333	0.5566996	0.5181451	
## 3	0.5000000	-2.000000	97.2500	0.5514663	0.5098640	
## 4	0.7500000	1.750000	104.5000	0.5502970	0.5075018	
## 5	1.0000000	20.000000	110.5000	0.5881799	0.5476244	
## 6	0.6666667	-5.333333	100.6667	0.5833257	0.5572799	
##	H2H_THREE_PT_RATE_B	H2H_REB_B	H2H_AST_B	H2H_TOV_B		
## 1	0.2965746	41.50000	23.75000	16.00000		
## 2	0.3594431	51.66667	23.66667	13.66667		
## 3	0.2190960	42.75000	20.50000	13.50000		
## 4	0.3061362	44.25000	20.25000	12.25000		
## 5	0.3506787	43.00000	27.00000	12.00000		
## 6	0.2698688	33.33333	18.66667	10.66667		

```

series_h2h_df <- series_h2h_df %>%
  mutate(
    H2H_WIN_PCT_DIFF =
      H2H_WIN_PCT_A - H2H_WIN_PCT_B,

    H2H_POINT_DIFF_DIFF =
      H2H_POINT_DIFF_A - H2H_POINT_DIFF_B,

    H2H_PTS_DIFF =
      H2H_PTS_A - H2H_PTS_B,

    H2H_TS_PCT_DIFF =
      H2H_TS_PCT_A - H2H_TS_PCT_B,

    H2H_EFG_PCT_DIFF =
      H2H_EFG_PCT_A - H2H_EFG_PCT_B,

    H2H_THREE_PT_RATE_DIFF =
      H2H_THREE_PT_RATE_A - H2H_THREE_PT_RATE_B,

    H2H_REB_DIFF =
      H2H_REB_A - H2H_REB_B,

    H2H_AST_DIFF =
      H2H_AST_A - H2H_AST_B,

    H2H_TOV_DIFF =
      H2H_TOV_A - H2H_TOV_B
  )

```

```

series_h2h_df <- series_h2h_df %>%
  select(
    SERIES_ID,
    season,
    TEAM_A,
    TEAM_B,
    WINNER,

    H2H_GAMES_A,
    H2H_GAMES_B,

    H2H_WIN_PCT_DIFF,
    H2H_POINT_DIFF_DIFF,
    H2H_PTS_DIFF,
    H2H_TS_PCT_DIFF,
    H2H_EFG_PCT_DIFF,
    H2H_THREE_PT_RATE_DIFF,
    H2H_REB_DIFF,
    H2H_AST_DIFF,
    H2H_TOV_DIFF
  )

```

```

head(series_h2h_df)

```

##		SERIES_ID	season	TEAM_A	
## 1	Atlanta Hawks vs Boston Celtics	2015-16		Atlanta Hawks	
## 2	Atlanta Hawks vs Cleveland Cavaliers	2015-16		Atlanta Hawks	
## 3	Charlotte Hornets vs Miami Heat	2015-16		Charlotte Hornets	
## 4	Cleveland Cavaliers vs Detroit Pistons	2015-16		Cleveland Cavaliers	
## 5	Cleveland Cavaliers vs Golden State Warriors	2015-16		Cleveland Cavaliers	
## 6	Cleveland Cavaliers vs Toronto Raptors	2015-16		Cleveland Cavaliers	
##	TEAM_B	WINNER	H2H_GAMES_A	H2H_GAMES_B	H2H_WIN_PCT_DIFF
## 1	Boston Celtics	1	4	4	0.5000000
## 2	Cleveland Cavaliers	0	3	3	-1.0000000
## 3	Miami Heat	0	4	4	0.0000000
## 4	Detroit Pistons	1	4	4	-0.5000000
## 5	Golden State Warriors	1	2	2	-1.0000000
## 6	Toronto Raptors	1	3	3	-0.3333333
##	H2H_POINT_DIFF_DIFF	H2H_PTS_DIFF	H2H_TS_PCT_DIFF	H2H_EFG_PCT_DIFF	
## 1	15.00000	7.500000	0.0754174671	0.0789715336	
## 2	-19.33333	-9.666667	-0.0535446125	-0.0589224873	
## 3	4.00000	2.000000	-0.0115507388	-0.0005714008	
## 4	-3.50000	-1.750000	0.0004321063	0.0072903298	
## 5	-40.00000	-20.000000	-0.1140160646	-0.1135847898	
## 6	10.66667	5.333333	0.0341855368	0.0251744796	
##	H2H_THREE_PT_RATE_DIFF	H2H_REB_DIFF	H2H_AST_DIFF	H2H_TOV_DIFF	
## 1	-0.005216925	-0.50	4.5000000	1.2500000	
## 2	-0.030710236	-9.00	-0.3333333	0.6666667	
## 3	0.091494226	-2.00	0.7500000	-3.2500000	
## 4	-0.005063862	-1.75	0.0000000	0.2500000	
## 5	-0.069407373	2.50	-10.0000000	1.5000000	
## 6	0.104774142	7.00	4.0000000	2.0000000	

```

h2h_correlations <- tibble(

  Variable = c(
    "H2H_WIN_PCT_DIFF",
    "H2H_POINT_DIFF_DIFF",
    "H2H_PTS_DIFF",
    "H2H_TS_PCT_DIFF",
    "H2H_EFG_PCT_DIFF",
    "H2H_THREE_PT_RATE_DIFF",
    "H2H_REB_DIFF",
    "H2H_AST_DIFF",
    "H2H_TOV_DIFF"
  ),

  Correlation = c(
    cor(series_h2h_df$H2H_WIN_PCT_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs"),

    cor(series_h2h_df$H2H_POINT_DIFF_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs"),

    cor(series_h2h_df$H2H_PTS_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs"),

    cor(series_h2h_df$H2H_TS_PCT_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs"),

    cor(series_h2h_df$H2H_EFG_PCT_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs"),

    cor(series_h2h_df$H2H_THREE_PT_RATE_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs"),

    cor(series_h2h_df$H2H_REB_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs"),

    cor(series_h2h_df$H2H_AST_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs"),

    cor(series_h2h_df$H2H_TOV_DIFF,
        series_h2h_df$WINNER,
        use = "complete.obs")
  )
)

h2h_correlations %>%
  arrange(desc(abs(Correlation)))

```



```
## # A tibble: 9 × 2
##   Variable          Correlation
##   <chr>             <dbl>
## 1 H2H_TS_PCT_DIFF    0.324
## 2 H2H_WIN_PCT_DIFF   0.323
## 3 H2H_EFG_PCT_DIFF   0.312
## 4 H2H_POINT_DIFF_DIFF 0.311
## 5 H2H_PTS_DIFF       0.311
## 6 H2H_AST_DIFF       0.215
## 7 H2H_REB_DIFF       0.148
## 8 H2H_TOV_DIFF       0.0548
## 9 H2H_THREE_PT_RATE_DIFF 0.0500
```

```
series_df <- read.csv("series_df.csv")
```

```
final_series_df <- series_df %>%
  left_join(
    series_h2h_df %>%
      select(
        SERIES_ID,
        season,
        H2H_WIN_PCT_DIFF,
        H2H_TS_PCT_DIFF,
        H2H_EFG_PCT_DIFF,
        H2H_POINT_DIFF_DIFF
      ),
    by = c("SERIES_ID", "season")
  )
```

```
head(final_series_df)
```

##		SERIES_ID	season	TEAM_A		
## 1	Atlanta Hawks vs Boston Celtics	2015-16		Atlanta Hawks		
## 2	Atlanta Hawks vs Cleveland Cavaliers	2015-16		Atlanta Hawks		
## 3	Charlotte Hornets vs Miami Heat	2015-16		Charlotte Hornets		
## 4	Cleveland Cavaliers vs Detroit Pistons	2015-16		Cleveland Cavaliers		
## 5	Cleveland Cavaliers vs Golden State Warriors	2015-16		Cleveland Cavaliers		
## 6	Cleveland Cavaliers vs Toronto Raptors	2015-16		Cleveland Cavaliers		
##	TEAM_B WINNER	TEAM_ID_A	TEAM_ID_B	WIN_PCT_DIFF		
## 1	Boston Celtics	1	1610612737 1610612738	0.000		
## 2	Cleveland Cavaliers	0	1610612737 1610612739	-0.110		
## 3	Miami Heat	0	1610612766 1610612748	0.000		
## 4	Detroit Pistons	1	1610612739 1610612765	0.158		
## 5	Golden State Warriors	1	1610612739 1610612744	-0.195		
## 6	Toronto Raptors	1	1610612739 1610612761	0.012		
##	NET_RATING_DIFF	TS_DIFF	THREE_PT_DIFF	OFF_RATING_DIFF	DEF_RATING_DIFF	
## 1	0.6	0.021	0.043278587	-1.2	-1.7	
## 2	-2.4	-0.006	-0.016515585	-5.6	-3.1	
## 3	1.0	0.000	0.127170795	0.9	0.0	
## 4	6.0	0.036	0.049406931	4.9	-1.0	
## 5	-4.3	-0.035	-0.009564644	-3.3	1.1	
## 6	1.9	0.006	0.065347599	0.9	-0.9	
##	EFG_DIFF	OREB_DIFF	TOV_DIFF	PLUS_MINUS_DIFF	SOS_DIFF	SRS_DIFF
## 1	0.028	-0.056	0.014	0.5	0.011000000	0.5110000
## 2	-0.008	-0.060	0.009	-2.5	0.017109756	-2.4828902
## 3	-0.006	-0.036	-0.019	0.9	-0.006707317	0.8932927
## 4	0.033	-0.019	0.003	5.9	-0.012012195	5.8879878
## 5	-0.039	0.027	-0.007	-4.4	-0.001329268	-4.4013293
## 6	0.020	-0.006	0.004	1.8	-0.002951220	1.7970488
##	H2H_WIN_PCT_DIFF	H2H_TS_PCT_DIFF	H2H_EFG_PCT_DIFF	H2H_POINT_DIFF_DIFF		
## 1	0.5000000	0.0754174671	0.0789715336	15.00000		
## 2	-1.0000000	-0.0535446125	-0.0589224873	-19.33333		
## 3	0.0000000	-0.0115507388	-0.0005714008	4.00000		
## 4	-0.5000000	0.0004321063	0.0072903298	-3.50000		
## 5	-1.0000000	-0.1140160646	-0.1135847898	-40.00000		
## 6	-0.3333333	0.0341855368	0.0251744796	10.66667		

```

predictor_df <- final_series_df %>%
  select(
    WIN_PCT_DIFF,
    TS_DIFF,
    EFG_DIFF,
    NET_RATING_DIFF,
    OFF_RATING_DIFF,

    H2H_WIN_PCT_DIFF,
    H2H_TS_PCT_DIFF,
    H2H_EFG_PCT_DIFF,
    H2H_POINT_DIFF_DIFF
  )

round(
  cor(predictor_df, use = "complete.obs"),
  2
)

```

##	WIN_PCT_DIFF	TS_DIFF	EFG_DIFF	NET_RATING_DIFF
## WIN_PCT_DIFF	1.00	0.64	0.68	0.94
## TS_DIFF	0.64	1.00	0.96	0.61
## EFG_DIFF	0.68	0.96	1.00	0.65
## NET_RATING_DIFF	0.94	0.61	0.65	1.00
## OFF_RATING_DIFF	0.69	0.77	0.75	0.70
## H2H_WIN_PCT_DIFF	0.56	0.35	0.38	0.54
## H2H_TS_PCT_DIFF	0.52	0.51	0.53	0.54
## H2H_EFG_PCT_DIFF	0.55	0.52	0.56	0.57
## H2H_POINT_DIFF_DIFF	0.58	0.37	0.41	0.62

##	OFF_RATING_DIFF	H2H_WIN_PCT_DIFF	H2H_TS_PCT_DIFF
## WIN_PCT_DIFF	0.69	0.56	0.52
## TS_DIFF	0.77	0.35	0.51
## EFG_DIFF	0.75	0.38	0.53
## NET_RATING_DIFF	0.70	0.54	0.54
## OFF_RATING_DIFF	1.00	0.43	0.44
## H2H_WIN_PCT_DIFF	0.43	1.00	0.70
## H2H_TS_PCT_DIFF	0.44	0.70	1.00
## H2H_EFG_PCT_DIFF	0.45	0.70	0.98
## H2H_POINT_DIFF_DIFF	0.46	0.81	0.85

##	H2H_EFG_PCT_DIFF	H2H_POINT_DIFF_DIFF
## WIN_PCT_DIFF	0.55	0.58
## TS_DIFF	0.52	0.37
## EFG_DIFF	0.56	0.41
## NET_RATING_DIFF	0.57	0.62
## OFF_RATING_DIFF	0.45	0.46
## H2H_WIN_PCT_DIFF	0.70	0.81
## H2H_TS_PCT_DIFF	0.98	0.85
## H2H_EFG_PCT_DIFF	1.00	0.84
## H2H_POINT_DIFF_DIFF	0.84	1.00

```

predictors <- final_series_df %>%
  select(
    WIN_PCT_DIFF,
    TS_DIFF,
    THREE_PT_DIFF,
    OFF_RATING_DIFF,
    H2H_TS_PCT_DIFF,

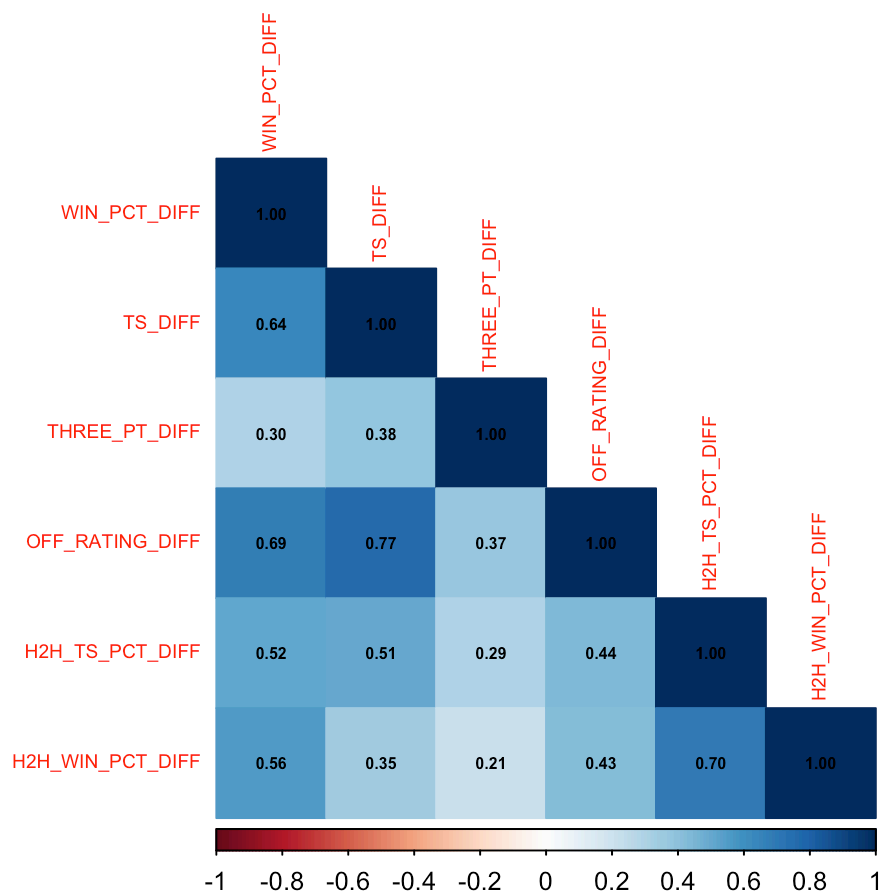
    H2H_WIN_PCT_DIFF
  )

```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
corrplot(
  cor(predictors, use = "complete.obs"),
  method = "color",
  type = "lower",
  addCoef.col = "black",
  number.cex = 0.5,
  tl.cex = 0.6
)
```



```
## Might not be necessary
library(usdm)
```

```
## Loading required package: terra
```

```
## Warning: package 'terra' was built under R version 4.4.3
```

```
## terra 1.9.27
```

```
##
## Attaching package: 'terra'
```

```
## The following object is masked from 'package:tidyr':
##
## extract
```

```
vif(predictors)
```

```
##           Variables      VIF
## 1    WIN_PCT_DIFF 2.417857
## 2         TS_DIFF 2.992732
## 3   THREE_PT_DIFF 1.207048
## 4  OFF_RATING_DIFF 2.984656
## 5   H2H_TS_PCT_DIFF 2.363242
## 6 H2H_WIN_PCT_DIFF 2.325431
```

```
## Probably not a good predictor
series_df <- series_df %>%
  mutate(
    HOME_COURT_WON =
      ifelse(WINNER == 1, 1, 0)
  )
```

```
mean(series_df$HOME_COURT_WON)
```

```
## [1] 0.5666667
```

we can do logistic regression with elastic net which will address multicollinearity in the data, we can safely do it if we are doing it for prediction